

Autocovariance of the Normalized Zeta-Process

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1 Notation and Definitions

Let $\zeta(s)$ denote the Riemann zeta function. Define the Riemann–Siegel theta function by

$$\vartheta(t) := \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi. \quad (1)$$

Let t_0 be the unique point where ϑ attains its global minimum on $(0, \infty)$, and set $\tau_0 := \vartheta(t_0)$. Let $\sigma : [\tau_0, \infty) \rightarrow [t_0, \infty)$ denote the inverse of ϑ restricted to $[t_0, \infty)$, so that $\vartheta(\sigma(\tau)) = \tau$.

The following asymptotic expansions hold as $t \rightarrow \infty$:

$$\vartheta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(t^{-1}), \quad (2)$$

$$\vartheta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(t^{-2}), \quad (3)$$

$$\vartheta''(t) = \frac{1}{2t} + O(t^{-3}). \quad (4)$$

Definition 1. Define:

$$V(T) := \int_0^T |\zeta(\tfrac{1}{2} + it)|^2 dt, \quad (5)$$

$$Y(\tau) := \frac{\zeta(\tfrac{1}{2} + i\sigma(\tau))}{\sqrt{\vartheta'(\sigma(\tau))}}, \quad (6)$$

$$W(\tau) := \int_{\tau_0}^{\tau} |Y(s)|^2 ds, \quad (7)$$

$$X(\tau) := Y(\tau) \sqrt{\frac{\tau}{W(\tau)}}. \quad (8)$$

We assume the following results from the companion document:

(F1) $W(\tau) = V(\sigma(\tau)) - V(\sigma(\tau_0))$, by the substitution $t = \sigma(s)$.

(F2) $W(\tau) = 2\tau(1 + \varepsilon(\tau))$ with $\varepsilon(\tau) = O(\frac{\log \log \tau}{\log \tau})$, obtained by combining (F1) with the Ingham–Atkinson mean-value theorem $V(T) = T \log \frac{T}{2\pi} + (2\gamma - 1)T + E(T)$, $E(T) = O(T^{1/2} \log^2 T)$, and the inversion $\sigma(\tau) = \frac{2\tau}{\log \tau} (1 + O(\frac{\log \log \tau}{\log \tau}))$.

(F3) $\lim_{T \rightarrow \infty} \frac{1}{T} \int_{\tau_0}^T |X(\tau)|^2 d\tau = 1$, proved via integration by parts on $\int \tau d(\log W)$ using only the pointwise asymptotic $\log W(\tau) = \log(2\tau) + o(1)$.

2 Statement of Results

Theorem 2. *For every fixed $h \in \mathbb{R}$, the Cesàro autocovariance*

$$R(h) := \lim_{T \rightarrow \infty} \frac{1}{T} \int_{\tau_0}^T X(\tau + h) \overline{X(\tau)} d\tau \quad (9)$$

exists and equals

$$R(h) = \frac{e^{-ih} \sin h}{h}, \quad (10)$$

with the convention $R(0) = 1$.

Corollary 3. *The spectral density of X is $\hat{S}(\omega) = \pi \cdot \mathbf{1}_{[-2,0]}(\omega)$.*

Proof of Corollary 3. Using the standard identity $\int_{-\infty}^{\infty} \frac{\sin h}{h} e^{-i\alpha h} dh = \pi \cdot \mathbf{1}_{[-1,1]}(\alpha)$:

$$\hat{S}(\omega) = \int_{-\infty}^{\infty} R(h) e^{-i\omega h} dh = \int_{-\infty}^{\infty} \frac{\sin h}{h} e^{-i(\omega+1)h} dh = \pi \cdot \mathbf{1}_{[-1,1]}(\omega + 1) = \pi \cdot \mathbf{1}_{[-2,0]}(\omega). \quad \square$$

3 Proof of Theorem 2

Separation of the Slowly Varying Prefactor

From the definitions (6) and (8),

$$X(\tau + h) \overline{X(\tau)} = Y(\tau + h) \overline{Y(\tau)} \cdot \frac{\sqrt{(\tau + h)\tau}}{\sqrt{W(\tau + h)W(\tau)}}. \quad (11)$$

Define the prefactor

$$\alpha(\tau, h) := \frac{\sqrt{(\tau + h)\tau}}{\sqrt{W(\tau + h)W(\tau)}}. \quad (12)$$

Lemma 4. *For each fixed $h \in \mathbb{R}$,*

$$\alpha(\tau, h) = \frac{1}{2} + O\left(\frac{\log \log \tau}{\log \tau}\right)$$

as $\tau \rightarrow \infty$.

Proof. By (F2), $W(\tau) = 2\tau(1 + \varepsilon(\tau))$ and $W(\tau + h) = 2(\tau + h)(1 + \varepsilon(\tau + h))$ with $\varepsilon(\tau), \varepsilon(\tau + h) = O(\log \log \tau / \log \tau)$. Then

$$\alpha(\tau, h) = \frac{\sqrt{(\tau + h)\tau}}{\sqrt{2(\tau + h)(1 + \varepsilon(\tau + h)) \cdot 2\tau(1 + \varepsilon(\tau))}} = \frac{1}{2} \cdot \frac{1}{\sqrt{(1 + \varepsilon(\tau + h))(1 + \varepsilon(\tau))}}.$$

Since $(1 + u)^{-1/2} = 1 - u/2 + O(u^2)$ for small u , the claim follows. $\square$

Write $\alpha(\tau, h) = \frac{1}{2} + \beta(\tau, h)$ with $\beta(\tau, h) = O(\log \log \tau / \log \tau)$. Then

$$\frac{1}{T} \int_{\tau_0}^T X(\tau + h) \overline{X(\tau)} d\tau = \frac{1}{2} \mathcal{I}(T, h) + \mathcal{E}(T, h), \quad (13)$$

where

$$\mathcal{I}(T, h) := \frac{1}{T} \int_{\tau_0}^T Y(\tau + h) \overline{Y(\tau)} d\tau, \quad \mathcal{E}(T, h) := \frac{1}{T} \int_{\tau_0}^T \beta(\tau, h) Y(\tau + h) \overline{Y(\tau)} d\tau.$$

Lemma 5. $\mathcal{E}(T, h) \rightarrow 0$ as $T \rightarrow \infty$ for every fixed h .

Proof. By Cauchy–Schwarz,

$$|\mathcal{E}(T, h)| \leq \sup_{\tau \geq \tau_0} |\beta(\tau, h)| \cdot \left(\frac{1}{T} \int_{\tau_0}^T |Y(\tau + h)|^2 d\tau \right)^{1/2} \left(\frac{1}{T} \int_{\tau_0}^T |Y(\tau)|^2 d\tau \right)^{1/2}.$$

By (F2), $\frac{1}{T} \int_{\tau_0}^T |Y(\tau)|^2 d\tau = W(T)/T = 2 + o(1)$. The shifted integral $\frac{1}{T} \int_{\tau_0}^T |Y(\tau + h)|^2 d\tau = (W(T+h) - W(h))/T = 2 + o(1)$ by the same estimate. Since $\sup |\beta(\tau, h)| = O(\log \log T / \log T) \rightarrow 0$, the product tends to zero. $\square$

By (13) and Lemma 5, to prove Theorem 2 it suffices to show

$$\lim_{T \rightarrow \infty} \mathcal{I}(T, h) = \int_{-2}^0 e^{i\omega h} d\omega = \frac{2e^{-ih} \sin h}{h}, \quad (14)$$

since then $R(h) = \frac{1}{2} \cdot \frac{2e^{-ih} \sin h}{h} = \frac{e^{-ih} \sin h}{h}$.

Change of Variables to the Height Line

Set $t = \sigma(\tau)$, so $\tau = \vartheta(t)$ and $d\tau = \vartheta'(t) dt$. Define the induced height-shift

$$\delta_h(t) := \sigma(\vartheta(t) + h) - t, \quad (15)$$

so that $Y(\tau + h)$ evaluated at $\tau = \vartheta(t)$ involves $\zeta(\frac{1}{2} + i(t + \delta_h(t)))$.

Lemma 6. For each fixed h , as $t \rightarrow \infty$,

$$\delta_h(t) = \frac{h}{\vartheta'(t)} + O\left(\frac{1}{t(\log t)^3}\right).$$

In particular, $\delta_h(t) \vartheta'(t) = h + O(1/(t(\log t)^2))$.

Proof. Taylor-expand σ at the point $\vartheta(t)$:

$$\sigma(\vartheta(t) + h) - \sigma(\vartheta(t)) = h \sigma'(\vartheta(t)) + \frac{h^2}{2} \sigma''(\xi)$$

for some ξ between $\vartheta(t)$ and $\vartheta(t) + h$. Since $\sigma'(\vartheta(t)) = 1/\vartheta'(t)$ and

$$\sigma''(s) = -\frac{\vartheta''(\sigma(s))}{(\vartheta'(\sigma(s)))^3},$$

the estimates (3) and (4) give $\sigma''(\xi) = O(1/(t(\log t)^3))$. The second claim follows by multiplying through by $\vartheta'(t)$. $\square$

Changing variables $\tau \rightarrow t$ in $\mathcal{I}$ and writing $T_* := \sigma(T)$ for the upper limit:

$$\mathcal{I}(T, h) = \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \frac{\zeta(\frac{1}{2} + i(t + \delta_h(t))) \overline{\zeta(\frac{1}{2} + it)}}{\sqrt{\vartheta'(t + \delta_h(t)) \vartheta'(t)}} \vartheta'(t) dt. \quad (16)$$

Lemma 7. For fixed h ,

$$\frac{\sqrt{\vartheta'(t)}}{\sqrt{\vartheta'(t + \delta_h(t))}} = 1 + O\left(\frac{1}{t \log t}\right).$$

Proof. Taylor-expand: $\vartheta'(t + \delta) = \vartheta'(t) + \delta \vartheta''(t) + O(\delta^2 \vartheta'''(t))$. Since $\delta = O(1/\log t)$, $\vartheta''(t) = O(t^{-1})$, and $\vartheta'''(t) = O(t^{-2})$:

$$\frac{\vartheta'(t + \delta)}{\vartheta'(t)} = 1 + O\left(\frac{1}{t \log t}\right).$$

Taking the inverse square root preserves the order. $\square$

Write the ratio in Lemma 7 as $1 + \gamma(t)$ with $\gamma(t) = O(1/(t \log t))$. After cancelling $\sqrt{\vartheta'(t)}$ from the numerator and denominator in (16):

$$\mathcal{I}(T, h) = \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \zeta\left(\frac{1}{2} + i(t + \delta_h(t))\right) \overline{\zeta\left(\frac{1}{2} + it\right)} dt + \mathcal{J}(T), \quad (17)$$

where

$$\mathcal{J}(T) = \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \gamma(t) \zeta\left(\frac{1}{2} + i(t + \delta_h(t))\right) \overline{\zeta\left(\frac{1}{2} + it\right)} dt.$$

By Cauchy–Schwarz and the mean-value estimate $\int_{t_0}^{T_*} |\zeta(\frac{1}{2} + it)|^2 dt = V(T_*) \sim T_* \log T_*$, together with $\vartheta(T_*) \asymp T_* \log T_*$ and $\sup_{t \geq t_0} |\gamma(t)| = O(1/(t_0 \log t_0)) \rightarrow 0$ as the lower integration limit is taken large:

$$|\mathcal{J}(T)| \leq \sup_{t \geq t_0} |\gamma(t)| \cdot \frac{V(T_*)^{1/2} \cdot V(T_* + |\delta|)^{1/2}}{\vartheta(T_*)} = o(1).$$

Therefore it suffices to evaluate

$$\lim_{T_* \rightarrow \infty} \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \zeta\left(\frac{1}{2} + i(t + \delta_h(t))\right) \overline{\zeta\left(\frac{1}{2} + it\right)} dt = \int_{-2}^0 e^{i\omega h} d\omega. \quad (18)$$

Expansion via the Approximate Functional Equation

For t sufficiently large, the Riemann–Siegel formula gives

$$\zeta\left(\frac{1}{2} + it\right) = A(t) + e^{-2i\vartheta(t)} \overline{A(t)} + \mathcal{R}(t), \quad (19)$$

where

$$A(t) := \sum_{n \leq N(t)} n^{-1/2 - it}, \quad N(t) := \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor, \quad (20)$$

and $\mathcal{R}(t) = O(t^{-1/4})$. (Here $\overline{A(t)} = \sum_{n \leq N(t)} n^{-1/2 + it} =: B(t)$.)

Abbreviate $\delta = \delta_h(t)$ throughout. The product $\zeta(\frac{1}{2} + i(t + \delta)) \overline{\zeta(\frac{1}{2} + it)}$ expands into four main terms plus cross-terms involving $\mathcal{R}$:

$$\text{I} = A(t + \delta) \overline{A(t)}, \quad (21)$$

$$\text{II} = A(t + \delta) e^{2i\vartheta(t)} B(t), \quad (22)$$

$$\text{III} = e^{-2i\vartheta(t + \delta)} B(t + \delta) \overline{A(t)}, \quad (23)$$

$$\text{IV} = e^{-2i\vartheta(t + \delta) + 2i\vartheta(t)} B(t + \delta) \overline{B(t)}. \quad (24)$$

Term I: Diagonal of the Upper Band

Expanding (21):

$$A(t + \delta) \overline{A(t)} = \sum_{m, n \leq N(t)} \frac{1}{\sqrt{mn}} (n/m)^{it} m^{-i\delta}. \quad (25)$$

(We use $N(t + \delta) = N(t)$ for large t since $\delta \rightarrow 0$; the finitely many terms where $N(t + \delta) \neq N(t)$ contribute $O(t^{-1/2})$ and are negligible.)

The diagonal $m = n$.

$$D_I(t) := \sum_{n \leq N(t)} \frac{n^{-i\delta_h(t)}}{n} = \sum_{n \leq N(t)} \frac{e^{-i\delta_h(t) \log n}}{n}. \quad (26)$$

By Lemma 6, $\delta_h(t) \log n = \frac{h \log n}{\vartheta'(t)} + O\left(\frac{\log n}{t(\log t)^3}\right)$. Define the spectral variable

$$\omega_n(t) := -\frac{\log n}{\vartheta'(t)}. \quad (27)$$

For $1 \leq n \leq N(t)$ with $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, one has $0 \leq \log n \leq \frac{1}{2} \log \frac{t}{2\pi}$, and by (3) $\vartheta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(t^{-2})$, so

$$\omega_n(t) \in [-1 - O(t^{-2}), 0]. \quad (28)$$

The phase in (26) is $-i\delta_h(t) \log n = i\omega_n(t) h (1 + O(1/(t \log^2 t)))$.

To evaluate the sum, apply the Abel summation identity. Set $L := \log N(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(1)$ and $f(u) := e^{-iuh}$. Then:

Lemma 8. *For a smooth function $f : [0, 1] \rightarrow \mathbb{C}$,*

$$\sum_{n \leq N} \frac{f\left(\frac{\log n}{L}\right)}{n} = L \int_0^1 f(u) du + O(\|f\|_\infty + \|f'\|_\infty),$$

where $L = \log N$.

Proof. Write $a(x) := \sum_{n \leq x} 1/n = \log x + \gamma + O(1/x)$. By Abel summation,

$$\sum_{n \leq N} \frac{f\left(\frac{\log n}{L}\right)}{n} = f(1) a(N) - \int_1^N a(x) \frac{d}{dx} \left[f\left(\frac{\log x}{L}\right) \right] dx.$$

Since $a(x) = \log x + \gamma + O(1/x)$ and $\frac{d}{dx} f\left(\frac{\log x}{L}\right) = \frac{f'(\log x/L)}{xL}$, the integral is

$$\int_1^N \frac{(\log x + \gamma) f'(\log x/L)}{xL} dx + O(1).$$

Substituting $u = \log x/L$ and using $\log x = uL$, $dx = xL du$:

$$= \int_0^1 (uL + \gamma) f'(u) du + O(1).$$

Integration by parts gives $\int_0^1 uL f'(u) du = L[uf(u)]_0^1 - L \int_0^1 f(u) du$, and $\int_0^1 \gamma f'(u) du = \gamma(f(1) - f(0))$. Combining with $f(1) a(N) = f(1)(L + \gamma + O(1/N))$:

$$\sum_{n \leq N} \frac{f(\log n/L)}{n} = L \int_0^1 f(u) du + O(\|f\|_\infty + \|f'\|_\infty). \quad \square$$

Apply Lemma 8 with $f(u) = e^{-iuh}$ (so $\|f\|_\infty = 1$, $\|f'\|_\infty = |h|$) and note $L = \vartheta'(t) + O(t^{-2})$:

$$D_I(t) = \vartheta'(t) \int_0^1 e^{-iuh} du + O(1) = \vartheta'(t) \int_{-1}^0 e^{i\omega h} d\omega + O(1), \quad (29)$$

where the last step is the substitution $\omega = -u$.

The off-diagonal $m \neq n$. The off-diagonal terms in (25) are

$$\text{OD}_I(t) = \sum_{\substack{m, n \leq N(t) \\ m \neq n}} \frac{1}{\sqrt{mn}} (n/m)^{it} m^{-i\delta_h(t)}.$$

For each pair (m, n) with $m \neq n$, the factor $(n/m)^{it} = e^{it \log(n/m)}$ oscillates at frequency $\log(n/m) \neq 0$ with respect to t . The factor $m^{-i\delta_h(t)} = e^{-i\delta_h(t) \log m}$ is slowly varying: since $\delta_h(t) = O(1/\log t)$ and $\log m \leq \frac{1}{2} \log t$, its derivative with respect to t is

$$\frac{d}{dt} [\delta_h(t) \log m] = \delta'_h(t) \log m = O\left(\frac{\log m}{t(\log t)^2}\right) = O\left(\frac{1}{t \log t}\right),$$

which is $o(1)$ relative to the principal oscillation frequency $|\log(n/m)| \geq \log(1 + 1/N(t)) \geq c/\sqrt{t}$.

To control the Cesàro average of OD_I , write $g_{m,n}(t) := m^{-i\delta_h(t)}/\sqrt{mn}$ and note $|g_{m,n}(t)| = 1/\sqrt{mn}$. By partial summation in t (integrating $e^{it \log(n/m)}$ and differentiating the slowly varying factor $g_{m,n}$), the integral $\int_{t_0}^{T_*} g_{m,n}(t) (n/m)^{it} dt$ is bounded by

$$\frac{2\|g_{m,n}\|_\infty}{|\log(n/m)|} + \frac{1}{|\log(n/m)|} \int_{t_0}^{T_*} |g'_{m,n}(t)| dt = O\left(\frac{1}{\sqrt{mn} |\log(n/m)|}\right) + O\left(\frac{T_*}{\sqrt{mn} |\log(n/m)| t_0 \log t_0}\right).$$

Summing over all $m \neq n$ with $m, n \leq N(T_*)$ and using the Montgomery–Vaughan bound $\sum_{m \neq n} \frac{1}{\sqrt{mn} |\log(n/m)|} \leq C \sum_{n \leq N} \frac{1}{n} \cdot \log \log N = O((\log N)(\log \log N))$ yields

$$\frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \text{OD}_I(t) dt = O\left(\frac{(\log T_*)(\log \log T_*)}{T_* \log T_*}\right) = o(1).$$

Combining:

$$\frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} I(t) dt = \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \vartheta'(t) dt \cdot \int_{-1}^0 e^{i\omega h} d\omega + o(1) = \int_{-1}^0 e^{i\omega h} d\omega + o(1), \quad (30)$$

since $\int_{t_0}^{T_*} \vartheta'(t) dt = \vartheta(T_*) - \vartheta(t_0) = \vartheta(T_*)(1 + o(1))$.

Term IV: Diagonal of the Lower Band

The phase prefactor in (24) is $e^{-2i\vartheta(t+\delta)+2i\vartheta(t)}$. By the mean-value theorem and Lemma 6:

$$\vartheta(t+\delta) - \vartheta(t) = \delta \vartheta'(t) + O(\delta^2 \vartheta''(t)) = h + O\left(\frac{1}{t(\log t)^2}\right). \quad (31)$$

So the prefactor is $e^{-2ih}(1 + O(1/(t(\log t)^2)))$.

The product $B(t+\delta) \overline{B(t)}$ expands as

$$B(t+\delta) \overline{B(t)} = \sum_{m, n \leq N(t)} \frac{1}{\sqrt{mn}} (m/n)^{it} m^{i\delta}.$$

(Note the signs: $B(t) = \sum n^{-1/2+it}$, so $\overline{B(t)} = \sum n^{-1/2-it}$.)

The diagonal $m = n$ contributes

$$D_{\text{IV}}(t) = e^{-2ih} \sum_{n \leq N(t)} \frac{e^{i\delta_h(t) \log n}}{n} \cdot (1 + O(1/(t(\log t)^2))).$$

The phase per summand is $e^{i\delta_h(t) \log n} = e^{-i\omega_n(t) h(1+o(1))}$, with ω_n as in (27) and the sign reversed relative to Term I. Including the prefactor e^{-2ih} , the total phase of the n -th term is $e^{-2ih} e^{-i\omega_n h} = e^{i(-\omega_n - 2)h}$.

Substituting $\tilde{\omega} = -\omega_n - 2$, which maps $\omega_n \in [-1, 0]$ to $\tilde{\omega} \in [-2, -1]$, and applying Lemma 8 in the same manner:

$$D_{\text{IV}}(t) = \vartheta'(t) \int_{-2}^{-1} e^{i\omega h} d\omega + O(1). \quad (32)$$

The off-diagonal terms of $B(t+\delta)\overline{B(t)}$ are controlled by the same partial-summation and Montgomery–Vaughan argument as for Term I (the roles of m and n are interchanged, but the structure is identical). Therefore:

$$\frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \text{IV}(t) dt = \int_{-2}^{-1} e^{i\omega h} d\omega + o(1). \quad (33)$$

Terms II and III: Rapidly Oscillating Cross-Terms

Term II in (22) carries the overall phase $e^{2i\vartheta(t)}$, and Term III carries $e^{-2i\vartheta(t+\delta)} = e^{-2i\vartheta(t)} e^{-2ih+o(1)}$. In both cases, the phase $e^{\pm 2i\vartheta(t)}$ has instantaneous frequency $\pm 2\vartheta'(t) \rightarrow \pm\infty$.

The amplitudes (products of Dirichlet polynomials of length $N(t)$) satisfy

$$\frac{1}{T_*} \int_{t_0}^{T_*} |A(t+\delta)| |B(t)| dt \leq \left(\frac{V(T_* + |\delta|)}{T_*} \right)^{1/2} \left(\frac{V(T_*)}{T_*} \right)^{1/2} = O(\log T_*).$$

Lemma 9. *Let $f : [t_0, \infty) \rightarrow \mathbb{C}$ satisfy $\frac{1}{T} \int_{t_0}^T |f(t)| dt = O((\log T)^A)$ for some $A > 0$, and let $\lambda(t)$ be a real-valued C^1 function with $\lambda'(t) \rightarrow \infty$ as $t \rightarrow \infty$. Then*

$$\frac{1}{\lambda(T)} \int_{t_0}^T f(t) e^{i\lambda(t)} dt \rightarrow 0.$$

Proof. For any $M > 0$, split the integral at the point t_M where $\lambda'(t) \geq M$ for all $t \geq t_M$. The interval $[t_0, t_M]$ contributes $O(1/\lambda(T)) = o(1)$. On $[t_M, T]$, integrate by parts:

$$\int_{t_M}^T f(t) e^{i\lambda(t)} dt = \left[\frac{f(t) e^{i\lambda(t)}}{i\lambda'(t)} \right]_{t_M}^T - \int_{t_M}^T \frac{d}{dt} \left(\frac{f(t)}{\lambda'(t)} \right) e^{i\lambda(t)} dt.$$

The boundary terms are $O(\|f\|/M)$. For the integral, the derivative $\frac{d}{dt}(f/\lambda')$ is integrable (since f grows at most polynomially and λ' grows), and dividing by $\lambda(T)$ gives a contribution that is $o(1)$ as $T \rightarrow \infty$ for any fixed M . Since M is arbitrary, the limit is zero. $\square$

Apply Lemma 9 with $\lambda(t) = \pm 2\vartheta(t)$ (so $\lambda'(t) = \pm 2\vartheta'(t) \rightarrow \pm\infty$) and f equal to the product of Dirichlet polynomials:

$$\frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \text{II}(t) dt \rightarrow 0, \quad \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \text{III}(t) dt \rightarrow 0. \quad (34)$$

Remainder Cross-Terms

Every cross-term involving $\mathcal{R}(t) = O(t^{-1/4})$ from the approximate functional equation (19) produces a product bounded pointwise by $O(t^{-1/4}) \cdot O(t^{1/4+\varepsilon}) = O(t^\varepsilon)$, where the second factor uses the convexity bound $\zeta(\frac{1}{2} + it) \ll t^{1/4+\varepsilon}$. After Cesàro averaging:

$$\frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} O(t^\varepsilon) dt = O\left(\frac{T_*^{1+\varepsilon}}{\vartheta(T_*)}\right) = O\left(\frac{T_*^\varepsilon}{\log T_*}\right) = o(1). \quad (35)$$

Assembly

Combining (30), (33), (34), and (35):

$$\begin{aligned} \lim_{T_* \rightarrow \infty} \frac{1}{\vartheta(T_*)} \int_{t_0}^{T_*} \zeta\left(\frac{1}{2} + i(t + \delta_h(t))\right) \overline{\zeta\left(\frac{1}{2} + it\right)} dt &= \int_{-1}^0 e^{i\omega h} d\omega + \int_{-2}^{-1} e^{i\omega h} d\omega \\ &= \int_{-2}^0 e^{i\omega h} d\omega. \end{aligned}$$

Computing the integral:

$$\int_{-2}^0 e^{i\omega h} d\omega = \frac{e^{i\omega h}}{ih} \Big|_{\omega=-2}^{\omega=0} = \frac{1 - e^{-2ih}}{ih} = \frac{2e^{-ih} \sin h}{h},$$

where the last equality uses $1 - e^{-2ih} = e^{-ih}(e^{ih} - e^{-ih}) = 2ie^{-ih} \sin h$.

By Lemma 7 and the identity (17), $\mathcal{I}(T, h)$ has the same limit. By (13) and Lemma 5:

$$R(h) = \frac{1}{2} \cdot \frac{2e^{-ih} \sin h}{h} = \frac{e^{-ih} \sin h}{h}.$$

At $h = 0$: $R(0) = \lim_{h \rightarrow 0} \frac{e^{-ih} \sin h}{h} = 1$, consistent with (F3). □